

WTW 158 Calculus

Assignment: 8 July 2020

When submitting this assignment, you agree to the following:

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, assignments, examinations and/or any other forms of assessment. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, assessments, assignments, examinations and/or any other forms of assessment.

TIME: 45 minutes to do the assignment, 30 minutes to prepare and submit

MARKS: 16

INSTRUCTIONS:

1. The assignment consists of 1 page and 4 questions.
2. Draw a line with a ruler at the end of each question.
3. Write your surname, initials and student number clearly in the beginning of your assignment.
4. **Show all your steps and calculations** and indicate where and how you apply theorems, where applicable.
5. You have 45 minutes to do the assignment.
6. You have 30 minutes to prepare the assignment for submittance and to submit.
7. Only handwritten assignments submitted in one PDF file will be accepted.
8. **Before submitting, make sure that you are satisfied with your answers and that you submit the correct document.**
9. No late assignments will be accepted.
10. No assignments sent via e-mail will be accepted.
11. No calculators may be used.

Question 1 [1,1,2]

Clearly indicate and explain what you do in the following:

- (a) Show that the vectors $\langle 6, 9, -3 \rangle$ and $\langle -4, -6, 2 \rangle$ are parallel.
- (b) Find a vector $\bar{a}$ (not the zero vector) that is orthogonal to $\langle -4, -6, 2 \rangle$. Show that $\bar{a}$ is indeed orthogonal to $\langle -4, -6, 2 \rangle$.
- (c) Find a unit vector in the direction opposite to the vector $\langle -4, -6, 2 \rangle$.

Question 2 [3]

Find the value(s) of x such that the angle between the vectors $\langle 0, 1, -1 \rangle$ and $\langle -1, x, 0 \rangle$ is $\frac{2\pi}{3}$. Show all your calculations.

Question 3 [2,1]

- (a) Sketch the region R enclosed by the curves $y = (x - 1)^2$ and $y = x + 1$, between $x = -1$ and $x = 2$. Indicate all relevant information on the sketch.
- (b) Write down, but do not evaluate, an integral to determine the area of the region R in Question 1(a).

Question 4 [4,2]

Consider the triangle with vertices $P(6, 2, 2)$, $Q(2, 0, -1)$ and $R(5, 1, -2)$.

- (a) Show that the triangle is a right triangle and indicate at which vertex is the right angle. Show all your steps.
- (b) Find the area of the triangle. Show all your calculations.